



Digital medicine

Bridging the chasm between AI and clinical implementation

Further reading

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Many advances in artificial intelligence (AI) for health care using deep neural networks have been commercialised. But few AI tools have been implemented in health systems. Why has this chasm occurred? Transparency, suitability, and adaptability are key reasons. The deployment of any new technology is usually managed centrally in hospitals and health systems. For the information technology (IT) teams, there is the concern that input data are drawn from outside the health setting and the algorithm performance, source code, and input data are unavailable to review. Many commercial AI applications are in radiology, but few are supported by evidence from published studies. And there are concerns that the algorithms were tested and validated using retrospective, in-silico data that may not reflect real-world clinical practice. Regulators reviewing a company's AI data are privy to considerable data, but these data are usually unavailable to health system IT teams or clinicians.

Equity, safety, and regulation are also crucial. When AI is imported from a commercial setting into a hospital or health system little is known about which data have been used to train the AI and there is uncertainty about ownership of the subsequent, real-patient data that AI draws on post implementation. Additionally, AI implementations can be hindered by inadequate information about the data used to make decisions and recommendations. Clinicians receiving AI-generated decisions rarely have oversight of the datapoints used to reach a specific decision, contributing to so-called algorithmic aversion. Such algorithmic aversion can also arise when it is unclear who (eg, developer companies, researchers, clinicians, or hospitals) should accept responsibility for the algorithm's decisions. Any consideration by hospital leadership to implement AI-based diagnostic and decision-making tools

should closely consider these concerns and barriers.

How can we bridge this chasm? Algorithmic aversion can be alleviated when an algorithm can be modified in the health setting or can learn from local data. Three steps will help optimise clinical use of AI. First, provide transparency about the datasets used for initial training of the AI tools. Second, enable the deconstruction of neural networks to make the features that drive the AI performance understandable for clinicians. Third, allow clinicians to retrain AI models with local data if the needs of their patients and hospital require it.

Progress is likely to come with the development of open-source AI trained on open data depositories and publicly shared algorithms. An example of the potential of this approach is InnerEye, a clinically validated AI for radiotherapy that has been implemented within a UK hospital and trained with hospital data and by hospital clinicians. If combined with privacy-preserving computing tools, such as federated learning, open-source AI could further remove barriers for the fast scalability of home-grown AI solutions developed in hospitals across the health system while maintaining clinician and patient trust in the ownership and regulation of data. Home-grown AI solutions, such as Mayo Clinic's AI-assisted screening tool for advanced review of the electrocardiogram and the rapid, AI-enabled genomics tool developed at Rady Children's Hospital for neonates with rare genetic diseases, point to how different prebuilt algorithmic tools and clinical expertise can come together to generate adapted AI to help address the specific needs of hospitals and patients. By mandating high standards for the implementation of open-source and locally adapted AI tools, their implementation could be rigorous yet flexible enough to fit within a hospital's unique IT infrastructure and governance frameworks.

Ultimately, it will be necessary for clinicians to embrace AI tools for substantial implementation to occur. That acceptance will be potentiated by rigorous clinical validation, a high bar not reflected by what companies are typically willing to pursue. Currently, algorithms are not further adapted after regulators' approval, which limits the autodidactic feature of deep learning that could be a persuasive tool for the value of an algorithm for IT staff and clinicians. The more that is done to address these multiple barriers, the sooner we will be able to see wider implementation of AI for routine patient care.

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